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AI Action Plan: The Arts and Technology and the Creative Industries: A Critical Driver for AI Innovation

The purpose of this response is to describe a targeted and specific focus that is essential to the nation's AI initiative. We anticipate many responses from other universities and industry that speak to the need for federal support for fundamental AI research, education, and workforce development. We fully support those recommendations and will not focus this response to duplicate consensus input. Rather, we write to advocate for the importance of the Arts in a National AI strategy to maintain US dominance in technology for the entertainment industry (music, motion pictures, video games, etc.; sectors that are growing at a rate often over 5% per annum); promote innovation and creativity in U.S. technology advances; and create the next generation of AI leaders though integration of AI and the arts. Current trends in AI development suggest that the U.S.'s key competitors in the technology space are unlikely to take advantage of the arts as a lever and driver for digital innovation and paradigm shifts. **A focus on the intersection of art and AI therefore represents a unique opportunity for the U.S. to maintain and advance its leading position in this field.**

"The more creativity we bring to AI, the more our AI can bring to creativity."
Jensen Huang, Founder, Nvidia Corporation.

The Arts[1] are already a strategic resource for the AI economy. AI and the arts are the drivers of areas such as AR/VR/xR, immersive and embodied simulation and training, and all aspects of UX design. Inexplicably, artists are often not at the table in setting direction for next generation AI.

[1] We understand the arts to include the fine and applied arts including but not limited to: visual art, time-based media, animation, design, architecture, drama, performance, music, creative writing and other creative practices.

Our response to this RFI is based on the following national challenge:

a) AI models and tools are centered on creative practices including creative writing, visual and aural composition, and industrial design and architecture;
b) as described, the arts bring a depth of knowledge and craft that will advance these tools and spur continued innovation as well as **emergent paradigm shifts** that have created companies from Meta (AI and social media) to Nvidia (AI and GPUs);
c) <u>there is an increased demand for creative cross-trained workers, entrepreneurs, and decision-makers who can meet the needs of an AI-ready workforce.</u>

Our competitors are not likely to recognize and value the role of the arts as national assets in the global contest, *but we can <u>and</u> should.*

THE NATIONAL CONTRIBUTION AND ECONOMIC IMPACT OF THE ARTS

"There are so many opportunities to do good work bringing the artistic perspective into business and technology. The future of innovation – and the future of humanity lies at this intersection of arts and technology."

Domhnaill Hernon, former Nokia Bell Labs Director in *Forbes* magazine (2020).

After WWII, the Massachusetts Institute of Technology (MIT) integrated the arts on campus to contextualize the "why" and "for whom" of new technologies to augment the "how." Former MIT president, Jerome B. Wiesner (1971-1980), noted that "[t]aken together, the arts, science and technology form a triple anvil on which to forge a new kind of apprenticeship for a complex world" (Jerome B. Wiesner President of MIT 1971-1980). Today's world is no less complex than the one Wiesner inhabited. One of the greatest sources of that complexity is the advent of AI.

Arts and design educational programs, colleges, and universities around the nation have driven the U.S.'s long history of innovation and experimentation in education, research, and creative practice. This is no less true with AI. Not only have we fostered innovations, but trailblazing graduates who are actively contributing to the future of AI in the world. As previously mentioned, arts graduates can be found in leading U.S. firms from Apple, Microsoft, Google, Meta, Amazon to NVIDIA Capitol One, Boeing, Lockhead Martin, SpaceX, and many, many, more. Our graduates are central to key sectors of American industry that collectively contribute more than $4 trillion to the U.S. economy. In particular:

- **Media and Entertainment:** spans motion picture, television, streaming, music, video, radio, books, and video gaming employed over 16 million workers in 2021 (trade.gov) These industries added approximately $2.9 trillion to the U.S. economy in 2021, accounting for 12.52% of the GDP. (trade.gov). This is projected to grow to $3.4 trillion by 2028, far exceeding the average GDP. Companies like NVidia
- **Creative Economies** which include gig work, freelance creatives, and individual entrepreneurs. In California, five core creative industries employed over 1 million full-time employees and over 400,000 contract workers. (dwt.com) Globally, the creative industries market was valued at approximately $2.78 trillion in 2023 and is projected to reach $4.06 trillion by 2032. (businessresearchinsights.com)
- **Architecture:** 127,300 architects / 105,847 licensed architects in the United States. (en.wikipedia.org). The broader architecture and engineering occupations employed about 195,000 individuals in 2023. (bls.gov). The US architectural services market was valued at approximately $73.01 billion in 2023 and is projected to grow at a compound annual growth rate (CAGR) of 4.2% from 2024 to 2030. (grandviewresearch.com)
- **Arts and Design:** There are an additional 267,200 graphic designers employed across various industries in the US (bls.gov). The U.S. industrial design market was valued at approximately $43.37 billion in 2023 and is projected to grow to $68.45 billion by 2032, exhibiting a compound annual growth rate (CAGR) of 5.20% during the forecast period. (marketresearchfuture.com). In 2022, the arts and cultural sector in the United States employed nearly 5.2 million workers. (arts.gov) The same year, this sector contributed approximately $1.1 trillion to the U.S. economy, accounting for 4.3% of the gross domestic product (GDP). (arts.gov)
- Our graduates are key contributors to the **Advanced Industries**, which employ around 12.3 million workers in the U.S. (brookings.edu). Their expertise in 3D printing, robotics, creative technologies, digital and interaction design, digital services and advanced manufacturing are central to success and continued innovation in AI.

At the Virginia Tech Institute for Creativity, Arts, and Technology (ICAT), our Science, Engineering, Arts, and Design (SEAD) teams have been innovating in immersive environments such as the Cube, to explore AI areas ranging from the previously mentioned fields of gaming, AR/VR/xR to Cybersecurity, US infrastructure development, smart agriculture, and cutting edge immersive audio-visual environments for simulation, training, and entertainment. ***This work pulls AI in***

directions that force unprecedented, emergent solutions that incremental AI development often misses.

By leveraging this connected knowledge, we are poised to innovate and lead by example. Holistic approaches to technology, that proactively consider potential risks and downstream impacts, are the most reliable route to economic growth and long-term technological dominance.

> "[Our] nation needs some of the most fiercely competitive and proudly autonomous global institutions in the United States to coalesce around the national interests of economic prosperity and economic security." (McLaughlin and Guile, *Issues in Science and Technology*, 9/22/21).

The arts are also essential to productively harnessing AI because they introduce teaching, research, and methodologies radically different from those already common in STEM. Closing the distance between STEM and the arts has exponential rewards, providing from the start "unfamiliar ways of thinking about concepts" (Stanich and Harp 2019), a catalyzing difference between modern historical modes of research and integrated approaches[2].

Here we summarize the main benefits of forging new partnerships between the arts and other fields leading in AI development and deployment:

1. ***For technology and business growth: Integrating the arts into the ways that we approach AI-related science, technology, and business will drive innovation and emergent paradigm shifts in these and other fields of inquiry.*** The divide between STEM and the arts is artificial and creates an impediment to national leadership.

[2] As explored by organizations such as the International Network of Science of Team Science (INSciTS); the National Science Foundation's (broader impacts criterion and its Advancing Research Impact on Society (ARIS); a2ru, the Center for Interdisciplinary Research, Collaboration, Learning, and Engagement (CIRCLE) at Michigan State University, among many others. On the industry side companies from Apple to Pixar to Electronic Arts to the Movie and Entertainment industry, have innovated because of the leadership of both artists and STEM employees. Therefore, for the US to lead the world in AI innovation, we must continue to advance this necessary partnership in ways that we can only imagine.

SEAD teams are needed to thrive in a hypercomplex world (Root-Bernstein, SCIENCE, July 2018). The postwar enthusiasm for the union of the arts and technology yielded MIT's Center for Advanced Visual Studies (CAVS), which served to "put technology in the hands of artists and to reinvigorate the creativity and human values of scientists and engineers" (Zacharias and Wisnioski, 2016). It also inspired the development of the remarkable Experiments in Art and Technology (E.A.T.) collective, which spawned numerous arts and technology projects around the world and is now integrated into Nokia Bell Labs. It is now nearly impossible to drive forward AI industry in almost all areas of transportation, healthcare, entertainment, etc., without SEAD teams.

2. ***For developing effective AI tools that enhance human abilities and produce useful and even transformative results.*** The workforce needs integrated knowledge to understand the context in which science and technology solutions must be implemented. Arts-integrated education shapes more well-rounded and more employable citizens, better equipped to solve wicked, contemporary problems. The arts introduce and strengthen skills in critical thinking, comfort in ambiguity, and novel approaches to problem-solving. Recently, experts at the Institute for the Future created a "transformation map" to analyze trends in AI based on discussions from the World Economic Forum in Davos. Their rubric showed that discussion trends mirrored many global AI concerns: generative AI+; bias and fairness in AI algorithms; operationalizing responsible AI; geopolitical impacts of AI, and AI and jobs. In a brief related to the project, one of the experts writes: "[I]t will be critical to involve people and experts from the most diverse backgrounds possible in guiding this technology in ways that enhance human capabilities and lead to positive outcomes." (Hollister, McGill University, 2025). The AI impacts and possibilities we face are complex, interdependent, and systemic, and our solutions must reflect this complexity. As we train future leaders to recognize and meet these trends and concerns, training in the arts offers perspectives that will be critical.

3. ***For humanity: Emphasizing human intelligence, creativity, and critical thinking that art and artists foster is and will be critical in the Age of AI.*** The practice and experience of the performing, visual, and design arts are deeply human and connective. As human-AI collaborations become more commonplace, harnessing the generative, creative nature of the arts lights the way to a better understanding of the

human role in AI. Recent convenings[3] of artists, engineers, technologists and experts in many disciplines explored how the arts are central to grappling with and leveraging the promise of AI. A report on the Creativity, Empathy, and AI summit at the National Academy of Science entitled `Centering the Arts in the Age of AI` summarizes in detail challenges and opportunities. The report documents an April 2024 summit co-convened by Virginia Tech, Carnegie Mellon, Penn State and the Alliance for the Arts in Research universities. These recommendations are based on the input of 65 delegates from 17 states, representing 18 universities, schools, and colleges, 5 industry organizations, and 9 non-profit organizations or government agencies. Recognizing these national needs, this coalition of higher education institutions have committed to shared action that will strengthen education and workforce development, create innovation and competition, and advance research and development of AI through the Arts. We are working to build pipelines and platforms to ensure our students are ready for the AI creative workforce. These efforts can be greatly strengthened through national and local policy.

CHALLENGES AND OPPORTUNITY: STRATEGIC INVESTMENTS IN ARTS & AI

Federal agencies have funded essential but limited work to meet the need of arts integration through the Research Labs program that emphasizes novel approaches to education, innovation and entrepreneurship. However, additional capacity is needed to fully realize the potential of arts graduates in AI. Cultivating partnerships and pipelines for educational research and innovations in workforce development this would help. The NSF's TIP directorate is positioned to support this, but prioritizes STEM research and activities, unintentionally excluding the important contributions that arts, design and creative industries bring.

[3] *Creativity, Empathy, and AI* (2024 Virginia Tech, Carnegie Mellon University, and Penn State University), *Generate I Integrate* (Rochester institute of Technology), and *Teaching the Arts in the Age of AI* (2025, University of Maryland)

POLICY NEEDS AND RECOMMENDATIONS:

The following policy recommendations and specific actions by the federal government will ensure American competitiveness in AI through the Arts. All of these actions have impacts that are clearly measurable:

A) Ensure the arts is represented in the national dialogue, decision making and funding of AI;

Action 1 - Engage Expert Advisors in the Arts: Compile a database of experts at leading research universities and in industry that have been collaborating at the nexus of arts, design, and technology. Utilize this expert pool to inform congressional caucuses and federally funded workforce development and research initiatives at NSF, DOD, and DOE. **Timeline: Immediate**

Action 2 - Foster Arts Representation in Funding and Policy: Incentivize and encourage the inclusion of expert advisors in the arts to ensure the representation of artists, designers, and creative practitioners in funding review panels, advisory decisions and policy decisions that concern AI and workforce development and AI and innovation. In particular, we recommend augmenting the NSF's TIP, CISE, and ENG advisory boards and their program's merit review panels with cross-sector arts expertise. **Timeline: Immediate**

B) Increase support for educational training and research in arts programs to ensure workforce readiness for AI;

Action 3 - **Dedicate Stronger Federal Funding Streams:** Allocate additional appropriations to strengthen NEA's grant programs in research, arts education, and media to fund practice and research at the intersection of arts and AI and add new programs to the NEA's portfolio that make strategic investments in innovation and creativity around AI and the Arts. This would 1) Build on current NEA efforts integrating arts methodologies a part of the National Science Foundation's Directorate for Technology, Innovation, and Partnerships (TIP); Directorate for Engineering (ENG); and Directorate for Computer and Information Sciences and Engineering (CISE), ; and 2) create greater awareness of ongoing NSF/NEA technology collaborations such as Algo Arts workshops and NSF funded ICC convenings at UCLA add new programs to the NEA's portfolio that make strategic investments in innovation and creativity around AI and the Arts. Build budget and incentivize non-arts focused funding agencies, like the NSF, DOD, and DOE to foster the role of the Arts in creating innovation across disciplines and application areas in AI. Allocate additional appropriations to strengthen federal grant programs in research, arts education, and media to fund practice and research at the intersection of arts and AI. Explore feasibility to add new

NEA programs that make strategic investments in innovation and creativity around AI and the Arts.4 Additionally, build budget and incentivize non-arts focused funding agencies, like the NSF, to foster the role of the Arts in creating innovation across disciplines in AI. To bolster this effort, connect and convene research institutions, federal agencies and entities such as CreativIT. **Timeline: 1 Year**

C) Incentivize cross-sector partnerships that strengthen K-16 training pipelines into industry and meet emerging needs for innovation through and with the arts.

Action 4 - Incentives for Arts+AI Partnerships: Provide a refundable tax credit for industry investment in regional arts partnerships to incentivize cross-sector and cross-institutional partnerships that strengthen K-16 pipelines for AI through and with the Arts. **Timeline: 2 Years**
Action 5 - Incentives for Workforce Development: *P*rovide a refundable tax credit for workforce development at the intersection of AI and the Arts, to include: recruiting interns from arts programs; artist-in-residency programs; fellowship programs that support artists, etc.) **Timeline: 2 Years**

D) Ensure innovation while protecting creative economies.

Action 6 - Mandate disclosure of AI Use for government-supported work: The government should be able to ask questions of AI's benefit to the American people, especially when incumbents and their technologies are supported with funding from the government [5]. We recommend making all appropriations for federal grant funding contingent on an AI disclosure. The NEA, NSF and other federal funding agencies should require that proposals discuss what forms of AI is used within the work, the implications for artists, designers, creative practitioners, and how responsible practices will be used and ensure that they protect artists' work and the economic prosperity of America's creative economies. This will give the government an actionable tool to safeguard innovation, as well as, the American creative economies. **Timeline: 2 Years**

[4] This would: 1) Build on current NEA efforts (integrating arts methodologies a part of the National Science Foundation's Directorate for Technology, Innovation, and Partnerships (TIP); and 2) create greater awareness of ongoing NSF/NEA technology collaborations such as Algorithmic Arts workshops and NSF funded Innovation, Culture, and Creativity convenings at UCLA add new programs to the NEA's portfolio that make strategic investments in innovation and creativity around AI and the Arts.

[5] "When a massive incumbent comes to us asking for safety regulations, we ought to ask whether that safety regulation is for the benefit of our people or for the benefit of the incumbent." JD Vance, AI Action Summit in Paris, France, on Tuesday, Feb. 11, 2025

Action 7 - *Promote AI Literacy*: As a public service, create ways to train students across all of the SEAD disciplines, citizens, and workers on ethical and productive use of AI tools and the tools themselves; and the useful role of AI in creative practice, research, writing, etc. in partnership with government, education and industry sectors.
Timeline: 3 Years

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